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FALL SEMESTER 2025-26

CSI3007 - Advanced Python Programming

SLOT: L7+L8+L27+L28

DAILY LAB ASSIGNMENT : LAB-17

Implementation of Sentiment Analysis and QR Code Generation using Python in Visual Studio Code

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3-point answer about using Visual Studio Code (VS Code) for Python:

1. Efficient Environment for Python Development

- VS Code provides an integrated and lightweight IDE for Python.
- It supports syntax highlighting, debugging, linting, virtual environments, and integrated terminal, making it easy to write, test, and run Python programs in one place.

2. TextBlob Integration

- With VS Code, you can easily use external Python libraries like **TextBlob** for **sentiment analysis, text processing, and natural language tasks**.
- Installing and running such libraries is straightforward using the **built-in terminal** (pip install textblob), and results are displayed instantly.

3. Built-in Extensions and Customization

- VS Code offers powerful **extensions** like **Python extension by Microsoft, Jupyter, and Pylance**, which enhance code completion, debugging, and visualization.
- You can also **customize themes, keybindings, and settings** to create a personalized development experience.

Aim: (QUESTION-1)

To perform **sentiment analysis** on a given text input using the **TextBlob** library in Python and classify the sentiment as **Positive, Negative, or Neutral**.

Prerequisites:

Before running the program, ensure the following packages are installed:

Python (version 3.8 or above)

TextBlob library

To install TextBlob, open the terminal in VS Code and run:

```
python -m pip install textblob
```

Procedure (in Visual Studio Code):

1. Open Visual Studio Code

Launch VS Code and create a new folder for your Python project.

2. Create a Python File

Go to File → New File

Save it as `sentiment.py` inside your folder.

3. Write the Python Code

Copy and paste the following program into your file.

4. Install Required Library

Open the integrated terminal (Ctrl + ~) and type:

```
python -m pip install textblob
```

5. Run the Program

In the terminal, type:

```
python sentiment.py
```

6. Enter Input

When prompted, enter a sentence and observe the polarity, subjectivity, and sentiment classification.

Code:

```
# Sentiment Analysis
```

```
from textblob import TextBlob
```

```
text = input("Enter a sentence: ")
```

```
blob = TextBlob(text)
```

```
print("Polarity:", blob.sentiment.polarity)
```

```
print("Subjectivity:", blob.sentiment.subjectivity)
```

```
if blob.sentiment.polarity > 0:
```

```
    print("Sentiment: Positive")
```

```
elif blob.sentiment.polarity < 0:
```

```
    print("Sentiment: Negative")
```

```
else:
```

```
    print("Sentiment: Neutral")
```

Output:

Example 1:

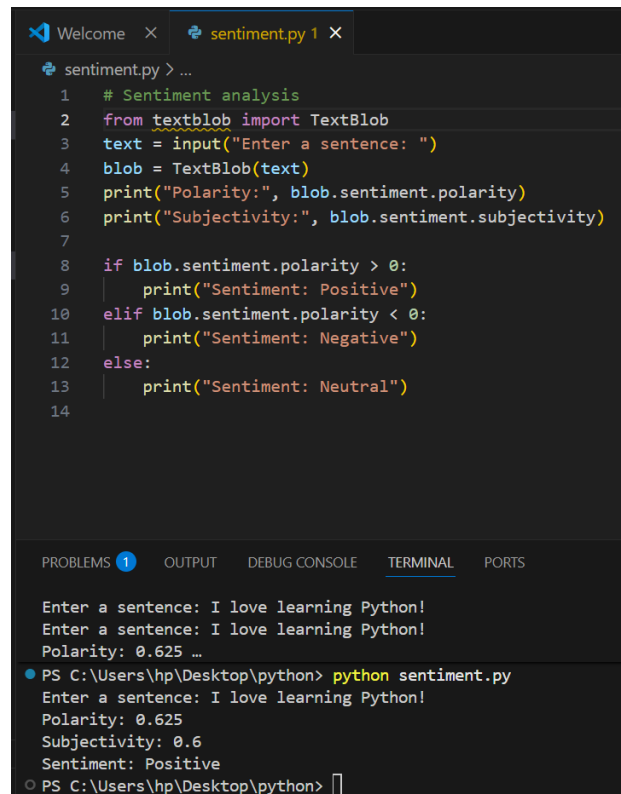
Enter a sentence: I love learning Python!

Polarity: 0.625

Subjectivity: 0.6

Sentiment: Positive

screenshots:



```

Welcome  X  sentiment.py 1 X
sentiment.py > ...
1  # Sentiment analysis
2  from textblob import TextBlob
3  text = input("Enter a sentence: ")
4  blob = TextBlob(text)
5  print("Polarity:", blob.sentiment.polarity)
6  print("Subjectivity:", blob.sentiment.subjectivity)
7
8  if blob.sentiment.polarity > 0:
9      print("Sentiment: Positive")
10 elif blob.sentiment.polarity < 0:
11     print("Sentiment: Negative")
12 else:
13     print("Sentiment: Neutral")
14
PROBLEMS 1  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Enter a sentence: I love learning Python!
Enter a sentence: I love learning Python!
Polarity: 0.625 ...
● PS C:\Users\hp\Desktop\python> python sentiment.py
Enter a sentence: I love learning Python!
Polarity: 0.625
Subjectivity: 0.6
Sentiment: Positive
○ PS C:\Users\hp\Desktop\python> 
```

Aim: (QUESTION-2)

To visualize student performance in different subjects using a **bar graph** with the help of the **Matplotlib** library in Python.

Prerequisites:

Before running the program, ensure the following are installed:

Python (version 3.8 or above)

Matplotlib library

To install Matplotlib, open the terminal in VS Code and type:

```
python -m pip install matplotlib
```

Code:

```
import matplotlib.pyplot as plt
subjects = ["Python", "C", "Java", "C++"]
scores = [88, 92, 79, 85]
plt.bar(subjects, scores, color='blue')
plt.title("Student Performance")
plt.xlabel("Subjects")
plt.ylabel("Marks")
plt.show()
```

Output:

Graph Display:

A **bar chart** is displayed with:

X-axis → Subjects (Python, C, Java, C++)

Y-axis → Marks (88, 92, 79, 85)

Blue bars representing student performance

Title: *Student Performance*

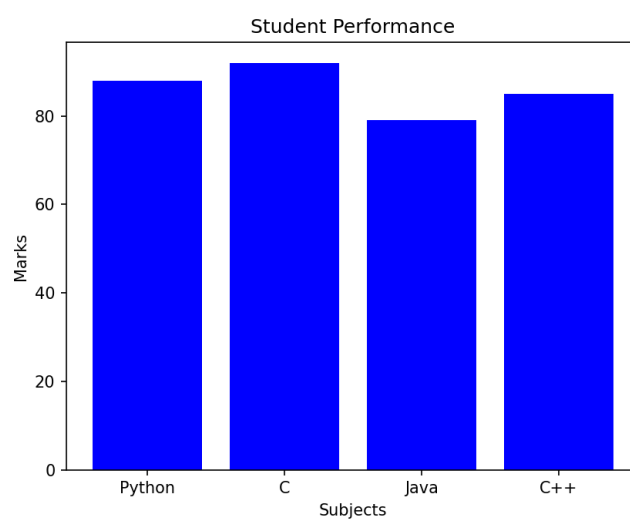
Observation:

From the graph, it is observed that the student scored the highest in **C (92)** and the lowest in **Java (79)**.

Screenshot:

```
Welcome | sentiment.py 1 | graph.py 1 X
graph.py > ...
1  import matplotlib.pyplot as plt
2
3  subjects = ["Python", "C", "Java", "C++"]
4  scores = [88, 92, 79, 85]
5
6  plt.bar(subjects, scores, color='blue')
7  plt.title("Student Performance")
8  plt.xlabel("Subjects")
9  plt.ylabel("Marks")
10 plt.show()
11
```

Figure 1



Aim: (QUESTION-3)

To generate a **QR Code** for a given URL or text using the **qrcode** library in Python and save it as an image file.

Prerequisites:

Before running the program, ensure the following packages are installed:

Python (version 3.8 or above)

qrcode and **Pillow (PIL)** libraries

Install them in the VS Code terminal using:

```
python -m pip install qrcode[pil]
```

The [pil] part ensures that the Pillow library (used for image generation) is also installed.

Code:

```
import qrcode
```

```
data = "https://www.google.com"
```

```
qr = qrcode.make(data)
```

```
qr.save("google_qr.png")
```

```
print("QR Code has been generated successfully!")
```

Output:

QR Code has been generated successfully!

```
VS Code Welcome | sentiment.py 1 | graph.py 1 | qr_generator.py 1 x
qr_generator.py > ...
1  import qrcode
2
3  data = "https://www.google.com"
4  qr = qrcode.make(data)
5  qr.save("google_qr.png")
6  print("QR Code has been generated successfully!")
7
```

